

Metodos matematicos da fisica: Homework 4

1 (pt). Show that there a bijection

$$\Phi: S^3 \rightarrow SU(2), (\alpha, \beta) \mapsto \begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \alpha \end{pmatrix}.$$

Here we view S^3 as the set of unit-length vectors on $\mathbb{C}^2$. **Remark:** This map is actually a homeomorphism. It follows that $SU(2)$ is a simply connected Lie group.